Contents

1. Introduction

The Balancing Tool is used to perform crankshaft balancing calculations based on a predefined Excel input file. It provides a simplified and more user-friendly interface for the existing calculation workflow. The tool reads the required engine, geometry, mass, and throw angle data from the uploaded file, executes the calculation, and displays the results in a structured format.

2. Required Input File

The tool requires an Excel input file containing a sheet named “Balancing”. The sheet must include the required input tables for engine configuration, crankshaft geometry, masses, and throw angles. The overall table structure must be maintained so that the tool can read the input data correctly.

2.1. Table 1- Engine configuration

Table 1: Engine configuration		
Description	Units	Value
Speed	rpm	1500
Number of cylinders	#	16
Cylinder arrangement	-	Vee
Vee angle	deg	90

Table 1 contains the basic engine configuration, including speed, number of cylinders, cylinder arrangement and vee angle. The correct units need to be included.

2.2. Table 2- Crankshaft geometry

Table 2: Crankshaft geometry		
Description	Units	Value
Stroke	mm	120
Conrod length	mm	411,5
Web CoG	mm	17,3
Counterweight CoG	mm	136,4
Distance between main journals	mm	320
Distance from main journal to bank A cylinder	mm	130
Distance from main journal to bank B cylinder	mm	190

Table 2 contains the main crankshaft geometry data required for the balancing calculation, such as stroke, conrod length, and relevant journal and cylinder distances.

2.3. Table 3- Rotating and oscillating masses

Table 3: Rotating and oscillating masses		
Description	Units	Value
Piston assembly mass	kg	11,93
Conrod oscillating mass (small end)	kg	9,86
Conrod rotating mass (big end)	kg	11,33
Crank pin mass	kg	8,662
Web mass	kg	14,45
Counterweight mass	kg	17,15

Table 3 contains the rotating and oscillating mass data, including piston mass, conrod masses, crank pin mass, web mass, and counterweight mass.

2.4. Table 4- Throw angles

Table 4: Throw angles								
Case number	Throw 1 (deg)	Throw 2 (deg)	Throw 3 (deg)	Throw 4 (deg)	Throw 5 (deg)	Throw 6 (deg)	Throw 7 (deg)	Throw 8 (deg)
8207	45	247,5	202,5	90	22,5	270	225	67,5
8200	45	247,5	202,5	90	270	22,5	67,5	225
17118	45	202,5	360	247,5	180	67,5	225	22,5
17485	45	225	180	0	247,5	67,5	382,5	202,5
7005	45	202,5	270	90	22,5	225	247,5	67,5
27156	45	202,5	337,5	157,5	270	90	225	22,5

Table 4 contains the crank throw angle combinations to be evaluated. Each row represents one case, and the corresponding throw angles define the crankshaft configuration for that case.

3. How to Use the Tool

To use the tool, upload the prepared Excel input file containing the required balancing data. After the file has been uploaded, the tool checks whether the expected sheet and table structure are available. Once the file is validated, click **Run Calculation** to start the balancing calculation. The results are then displayed in separate tabs for overview and detailed harmonic results.

4. Results an Calculation

Formeln hier noch einfügen